

**Time:** 5 minutes**Outcome:** Analyze Natural Monopoly & Regulation.**Difficulty:**

●●○ Intermediate

Natural Monopoly & Regulation



THE CORE IDEA

A natural monopoly has declining average cost across the relevant market range, making one large producer less costly than several small firms. The subsidy addresses the financing gap created by efficient pricing when scale economies keep ATC above MC. Average-cost pricing is a compromise between cost recovery and the $P=MC$ efficiency benchmark. Marginal-cost pricing can require financial support when MC lies below ATC.



HOW TO RECOGNIZE IT

- Look for average cost that continues falling across the relevant range of market demand.
- Marginal-cost pricing expands output toward efficiency but may leave the firm unable to cover average cost.
- Average-cost pricing can preserve financial viability but produces less than the efficient quantity.
- Evaluate regulation by comparing price, output, efficiency, and the need for subsidy or allowed profit.

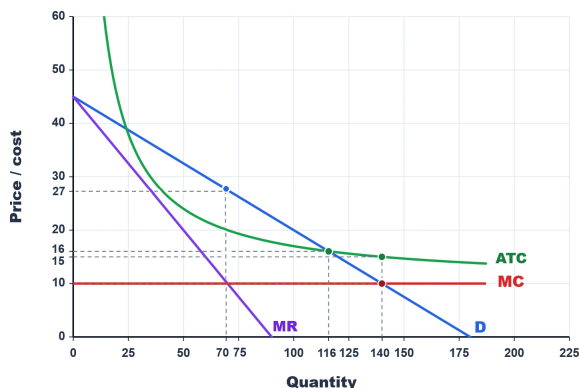


WATCH OUT

Do not confuse allocative efficiency with break even. Marginal-cost pricing can require financial support when MC lies below ATC. Natural monopoly is rooted in cost structure and economies of scale rather than legal exclusion alone.



WORKED EXAMPLE: REGULATING A NATURAL MONOPOLY



Marginal-cost pricing sets demand equal to MC, which the graph shows at 140 units and \$10. ATC there is about \$15, so the regulated firm would lose about $(\$15 - \$10) \times 140 = \$700$. Average-cost pricing near 115 units and \$16 avoids that loss but leaves price above marginal cost.



CHECK YOURSELF

What is the main compromise under average-cost pricing?

**READY?**

Return to the game and master this concept.